

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
LABORATORY OBSERVATION & RECORD

Course Name: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 01

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Branch & Section: CSM-B

Date of Execution: 25/08/2026

1. TITLE

Creating a Responsive User Registration Form / Static Web Page

2. AIM

To design and implement a responsive static web page with validation using HTML5, CSS3, and JavaScript incorporating semantic elements, inline styling, hyperlinking, and custom form controls.

3. OBJECTIVE

- To understand HTML5 document structure and semantic tags.
- To apply inline CSS styling for layout and text formatting.
- To construct web forms with HTML5 validation and form buttons.

4. THEORY

HTML5 introduced semantic elements like <section>, <header>, <main>, <article>, and <footer> which provide visual and structural meaning to web content. Form controls such as <fieldset>, <legend>, <input>, <select>, and <textarea> are used to capture user inputs cleanly. Attributes like 'required' allow browser-level input validation before submitting form data.

5. ALGORITHM / PROCEDURE

1. Initialize HTML5 document structure with head metadata and title.
2. Add header with navigation links (<nav>, <a>) and hero section (<section>).
3. Construct main page layout containing <main>, <article>, <form>, and <aside> sidebars.
4. Build an interactive form using <fieldset>, <legend>, text inputs, email inputs, dropdown menu (<select>), and action buttons.
5. Add document references list (,) and footer content (<footer>).
6. Save file, open in browser, and test form fields and links.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>
  <!-- Header Section -->
  <header>
    <table width="100%" border="0" cellspacing="0" cellpadding="10">
      <tr>
        <td>
          <h1>StaticWeb</h1>
        </td>
        <td align="right">
          <nav>
            <a href="#home">Home</a> &nbsp;|&nbsp;&nbsp;
            <a href="#articles">Articles</a> &nbsp;|&nbsp;&nbsp;
            <a href="#about">About</a> &nbsp;|&nbsp;&nbsp;
            <a href="#contact">Contact Us</a>
          </nav>
        </td>
      </tr>
    </table>
  </header>
  <hr>
  <!-- Hero / Showcase Section -->
  <section id="home">
    <center>
      <h2>Semantic Structures Without CSS</h2>
      <p>This layout uses raw HTML components to build an organized document
herachy that browsers can cleanly parse out-of-the-box.</p>
      <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
    </center>
  </section>
  <hr>
  <!-- Main Content Layout using a Structural Table -->
  <table width="100%" border="0" cellspacing="0" cellpadding="20">
    <tr valign="top">
      <!-- Left Side: Main Content Column -->
      <td width="70%">
        <main id="articles">
          <article>
            <h2>1. Core Structural Tags</h2>
            <p>Published: July 7, 2026</p>
            <p>When you omit cascading style sheets entirely, the
semantic markup handles all of the structural heavy lifting. Elements like
<code>&lt;main></code>, <code>&lt;article></code>, and
<code>&lt;section></code> tell screen readers and web scrapers exactly how
information flows.</p>
            <p>Using raw browser defaults ensures excellent loading
performance, perfect accessibility compliance, and absolute structural clarity.</p>
          </article>
          <br><hr><br>
          <article>
            <h2>2. Native Interaction Elements</h2>
            <p>Published: July 5, 2026</p>
            <p>Browsers come equipped with excellent interactive
elements by default. We can capture text inputs, provide multiple-choice
configurations, and submit structured information to backend parsers using pure
markup.</p>

```

```

        <h3>Interactive Demonstration Forms</h3>
        <form id="contact" action="#" method="post">
            <fieldset>
                <legend><b>User Inquiry Form</b></legend>
                <p>
                    <label for="name">Full Name: </label>
                    <input type="text" id="name" name="username"
required placeholder="John Doe">
                </p>
                <p>
                    <label for="email">Email Address: </label>
                    <input type="email" id="email" name="useremail"
required>
                </p>
                <p>
                    <label for="topic">Inquiry Topic: </label>
                    <select id="topic" name="usertopic">
                        <option value="general">General
Feedback</option>
                        <option value="technical">Technical
Architecture</option>
                        <option value="academics">Academic
Curriculum</option>
                    </select>
                </p>
                <p>
                    <label for="message">Message
Details:</label><br>
                    <textarea id="message" name="usermessage"
rows="5" cols="40" placeholder="Type your notes here..."></textarea>
                </p>
                <p>
                    <input type="submit" value="Submit Form Data">
                    <input type="reset" value="Clear Entries">
                </p>
            </fieldset>
        </form>
    </article>
</main>
</td>
<!-- Right Side: Sidebar Column -->
<td width="30%" bgcolor="#f4f4f4">
    <aside id="about">
        <h3>Document References</h3>
        <p>Static pages map data out directly without real-time
database queries or processor overhead.</p>
        <h4>Core Benefits:</h4>
        <ul>
            <li>Instant browser painting</li>
            <li>Low memory footprints</li>
            <li>High data compatibility</li>
        </ul>
        <h4>Required Components:</h4>
        <ol>
            <li>Document Type Definition</li>
            <li>Semantic Wrapper Blocks</li>
            <li>Data Input Control Interfaces</li>
        </ol>
    </aside>
</td>
</tr>
</table>
<hr>

```

```

        <!-- Footer Section -->
        <footer>
            <center>
                <p>&copy; 2026 Pure HTML Architecture. Assembled completely without
style configurations.</p>
            </center>
        </footer>
    </body>
</html>

```

7. CODE EXPLANATION

- <header>, <main>, <article>, <aside>, <footer>: Defines semantic page regions.
- <nav> & <a>: Defines internal section anchor links for smooth navigation.
- <fieldset> & <legend>: Groups form fields with a bold container header.
- <select> & <option>: Creates a native drop-down menu for topic selection.
- <textarea>: Provides a multi-line input box for messages.

8. TEST CASES AND EXECUTION DATA

1. TC-01: Page Load -> Renders navigation header, articles, sidebar, and form -> Passed
2. TC-02: Nav Links -> Anchor tags scroll to respective section IDs -> Passed
3. TC-03: Form Validation -> Required fields prevent empty submission -> Passed
4. TC-04: Reset Button -> Clears all entered inputs -> Passed

9. OUTPUT

GitHub Link: [https://github.com/hv455253-cmyk/FSWD-Lab](https://github.com/hv455253-cmyk/FSWD-Lab/blob/main/Record/week-1-record/week-1.html)

/blob/main/Record/week-1-

record/week-1.html

StaticWeb

[Home](#) | [Articles](#) | [About](#) | [Contact Us](#)

Semantic Structures Without CSS

This layout uses raw HTML5 components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.

[Read Our Articles Below ↓](#)

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like <main>, <article>, and <section> tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend servers using pure markup.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic: General Feedback ▼

Message Details:

Type your notes here....

Submit Form Data

Clear Entries

Document References

Static pages may data out directly without real-time database queries or processor overhead.

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

- Document Type Definition
- Semantic Wrapper Blocks
- Data Input Control Interfaces

© 2026 Pure HTML5 Architecture. Assembled completely without style configurations.

10. RESULT

A responsive static web page with form controls and semantic layout was successfully designed, executed, and verified using HTML5.

RUBRIC EVALUATION (Faculty Use Only)

Design & Logic (5 Marks): _____

Execution & Output (5 Marks): _____

Viva & Record (5 Marks): _____

TOTAL MARKS: [/ 15]

Faculty Signature: _____